

§1

THE SITUATION

Every kilometre a soldier carries ammunition, water or a wounded man is a kilometre someone can be killed on. Ukraine proved a machine can take that risk instead — over a hundred thousand times in 2026 alone, until ground robots were carrying eighty per cent of one brigade's logistics. None of those machines work here. Our ground is closed canopy that removes the link, the satellite fix and the light; heat that throttles the compute; salt, condensation and fungus that eat the hardware before anyone notices. The machines that would keep our people out of the open do not exist, and nobody outside this region has any reason to build them for us.

PROBLEM STATEMENT

Ideate a machine that takes a dangerous job off a person in Singapore's terrain — jungle, mud, coast or tidal ground. What it does is your decision. Showcase a proof of concept that makes the mechanism believable: a bench rig, a simulation, or one subsystem demonstrated.

§2

THINGS TO TAKE NOTE OF

01

NAVIGATION WITHOUT SKY OR LIGHT

Under canopy there is no sky, so no satellite fix and no satellite communications. One to two per cent of open-sky light reaches the floor, and the scene is self-similar green moving in wind, so feature-tracking drifts badly. Structure — trunks, buttress flare, terrain relief — is the only stable reference, which favours lidar and radar over anything vision-first.

02

SIGNAL ATTENUATION THROUGH WET VEGETATION

Wet vegetation attenuates 2.4 and 5.8 GHz severely, and worst in rain, which is when the link is most needed. Because our distances are short, a fibre tether becomes viable here in a way it never is across a twelve-hundred-kilometre front. Look for what is impractical at continental scale and obvious at ours.

03

RUGGEDISATION IN A CHALLENGING ENVIRONMENT

Salt aerosol, daily condensation and fungal growth attack connectors, enclosures and optics. Fungus has its own test method in MIL-STD-810. These present as an intermittent fault three weeks later, so screening matters more here than field repair.

04

OPERATOR-TO-PLATFORM RATIO

Ukrainian platforms are operator-centric. Fielding thousands is meaningless if each needs somebody watching it. Attack the ratio, not the platform.

05 ATTRITION AND RECOVERY

Ground robots are targets: Russian losses stood at 2,190 by August 2026. If it fails in a bad place, does a person walk in and retrieve it? That question eliminates more concepts than any technical limit.

06 MANGROVE, MUDFLAT AND TIDAL TRANSITION

Mangrove, mudflat and tidal transition is neither land nor water. A ground platform bogs; a boat grounds. Whether one platform crosses it without a change of mode is not known.

§3 TWO EXAMPLES

ONE — FLUIDISE THE GROUND RATHER THAN FIGHT IT

Saturated mud is neither solid nor liquid, and traction is the wrong tool against it. A mudskipper crosses it by oscillating its tail to fluidise the substrate underneath itself, and a robot built on that principle gained 67 per cent in speed and shed 46 per cent of its body drag on deformable granular terrain. The move is not a better track or a wider wheel. It is changing the state of the ground for the moment you need to cross it — and it is the kind of answer that only appears if you study what already lives there.

TWO — PUT THE EFFECTOR WHERE AN AIRCRAFT CANNOT GO

Under closed canopy an interceptor cannot fly, and a jammer cannot touch a fibre-optic drone. Ukraine's answer was a ground robot carrying an AI gun turret: it detects the drone, tracks it, calculates where it will be when the rounds arrive, and asks a human for permission to fire, engaging small drones out to several hundred metres. Related systems mount a 12.7 mm turret on a tracked chassis with an engagement range of about a kilometre. The move is recognising that the effector does not have to be airborne, and that a machine can hold the position a soldier would otherwise have to hold.